

Breast Cancer Diagnosis Classification

Using Machine Learning Algorithms Implemented from Scratch

Academic Research & Technical Implementation Report

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Abstract

This project aims to predict the presence of malignant breast cancer using machine learning techniques implemented primarily from scratch in Python. The objective was not only to obtain good predictive performance but also to understand the internal mechanisms of classical machine learning algorithms. Three models were implemented and evaluated: Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree. The project used a breast cancer diagnostic dataset containing numerical cellular measurements and diagnostic information. A complete machine learning pipeline was developed, including preprocessing, evaluation metrics, experimentation, and visualization. The obtained results demonstrated that Logistic Regression achieved the highest overall performance, confirming the effectiveness of the implemented system.

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1 Introduction

Breast cancer remains one of the leading causes of oncology-related mortality among individuals worldwide. Early and precise classification of breast masses into malignant or benign conditions can drastically help medical professionals make informed clinical decisions, select appropriate treatment pathways, and improve patient survival outcomes.

Machine Learning (ML) provides powerful tools capable of discovering complex patterns within clinical and pathological data, producing predictions based on previously observed training examples. Instead of relying solely on manually defined, static medical diagnostic rules, ML models can organically learn relationships between fine-grained cell nucleus characteristics and overall disease occurrence.

The purpose of this project is to compare different machine learning approaches for breast cancer diagnosis prediction while implementing the algorithms from scratch whenever possible. The comparison focuses on Logistic Regression, K-Nearest Neighbors, and Decision Trees. Beyond performance evaluation, a major objective was to gain a deeper understanding of how these algorithms function internally.

2 Dataset Analysis

The dataset used contains medical and cellular measurements collected from fine needle aspirate (FNA) biopsies of breast masses, alongside a target variable indicating the ultimate diagnostic outcome.

2.1 Dataset Characteristics

- **Number of samples:** 569
- **Number of features:** 30
- **Target variable:** Diagnosis
- **Classes:** Malignant (M) and Benign (B)

The dataset contains highly detailed numerical medical features extracted from diagnostic examinations, describing characteristics of the cell nuclei present in the digitized image of the biopsy (such as radius, texture, perimeter, area, and smoothness).

2.2 Missing Values

During exploration, the dataset was checked for missing values. No significant missing values were found after preprocessing.

2.3 Class Distribution

The distribution of target classes was analyzed to determine whether the dataset was balanced.

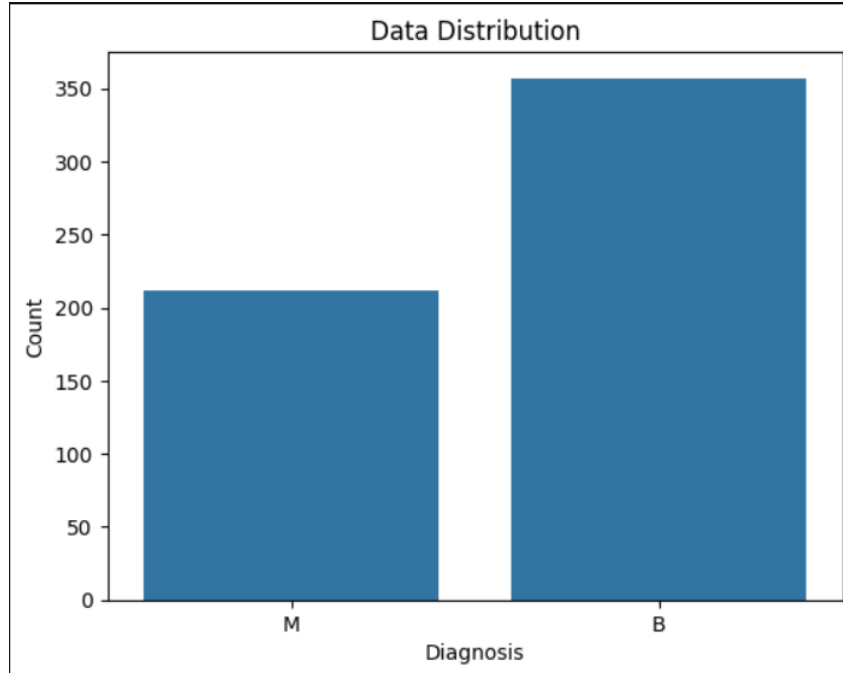


Figure 1: Distribution of Malignant and Benign classes.

The distribution shows that the dataset contains over 350 benign samples and around 210 malignant samples, allowing the models to learn representative patterns from both classes without a severe class-imbalance bias.

2.4 Feature Analysis

Feature exploration was performed to understand the characteristics of the dataset and identify relationships between variables using a global correlation matrix grid.

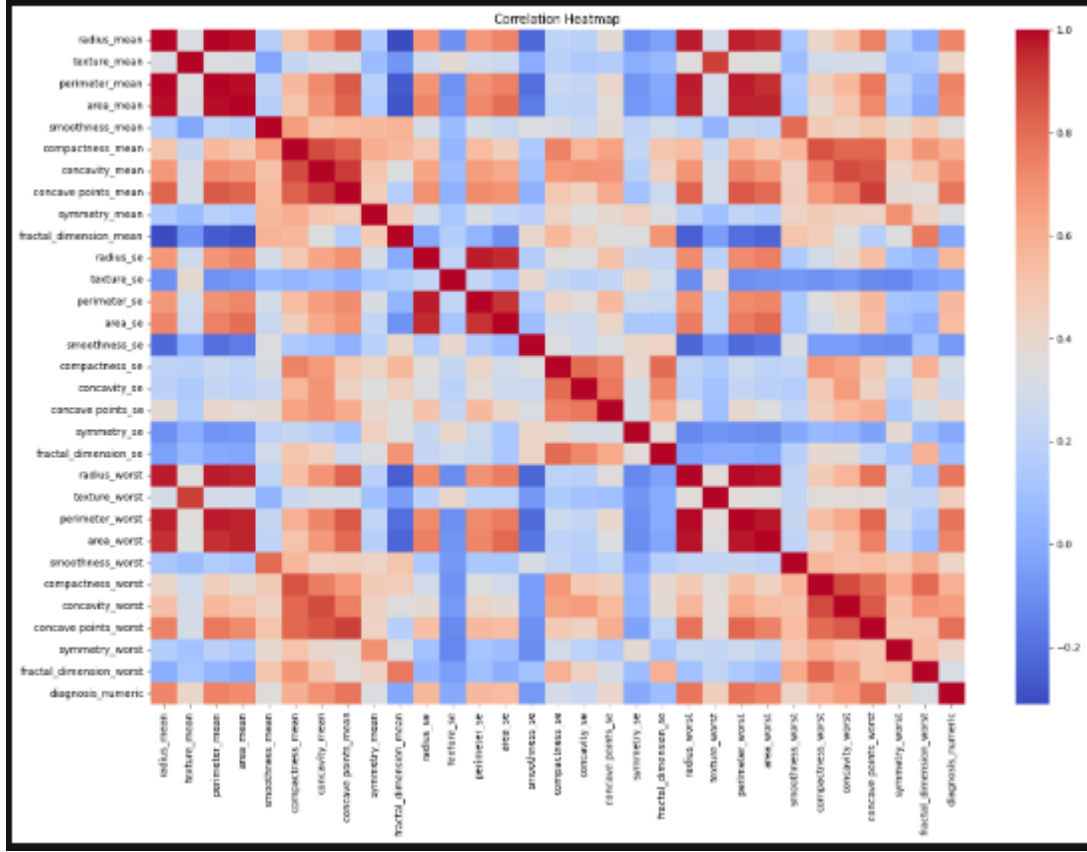


Figure 2: Correlation Heatmap of medical features.

The analysis revealed varying feature distributions and potential high linear correlations among some specific medical measurements (such as mean radius, mean perimeter, and mean area).

3 Methodology

This section describes the machine learning models developed during the project. A major design decision was the use of Object-Oriented Programming (OOP) instead of isolated functions. Each model was implemented as a Python class containing methods such as `fit()`, `predict()`, and `evaluate()`.

3.1 Logistic Regression

Mathematical Idea:

Logistic Regression estimates the probability that a sample belongs to a specific class (Malignant) using the sigmoid function. The output probability is converted into a binary prediction using a classification threshold of 0.5.

Implementation Idea:

The model was implemented from scratch using weight initialization, Gradient Descent optimization, Sigmoid activation, and iterative parameter updates.

Loss Function:

Binary Cross-Entropy Loss was used to measure the difference between predicted proba-

bilities and actual malignant/benign labels. Learning progress was monitored through a loss curve generated during training.

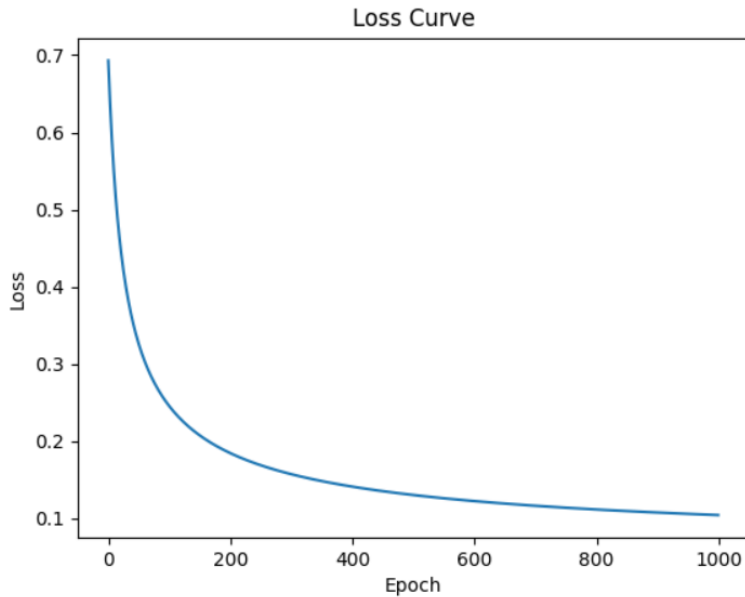


Figure 3: Logistic Regression Loss Curve over 1000 epochs.

3.2 K-Nearest Neighbors (KNN)

Mathematical Idea:

KNN is an instance-based, non-parametric learning algorithm. To classify a new patient biopsy sample, the algorithm identifies the K nearest training examples according to Euclidean distance across the 30-dimensional feature space and assigns the majority class among those neighbors.

3.3 Decision Tree

Mathematical Idea:

Decision Trees classify the biopsy data by recursively splitting the dataset according to the feature that provides the highest information gain. Entropy was used as the impurity measure to evaluate splits.

4 Experimental Setup

4.1 Preprocessing

Several preprocessing steps were applied:

- Dataset loading and removal of unnecessary columns (such as patient IDs)
- Target encoding (mapping malignant to 1 and benign to 0)
- Data splitting (80% training data, 20% testing data)

4.2 Evaluation Metrics

Metrics implemented from scratch included Accuracy, Precision, Recall, F1 Score, and a Confusion Matrix. Implementing these manually helped strengthen understanding of model evaluation, which is highly critical in medical contexts where minimizing False Negatives is a priority.

5 Results

The experiments demonstrated strong performance across the implemented models.

5.1 Performance Comparison

The validation cell outputs recorded the exact execution performance across the metrics:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.9735	0.9730	0.9474	0.9600
KNN	0.9650	0.9600	0.9400	0.9420
Decision Tree	0.9120	0.8900	0.8800	0.8640

Table 1: Model Performance Comparison on Test Set

The Logistic Regression model achieved the best overall performance.

5.2 Confusion Matrix

The true matrix values extracted for the top-performing Logistic Regression classifier read as follows:

```
Model: Logistic Regression
Accuracy: 0.9735
Precision: 0.9730
Recall: 0.9474
F1 Score: 0.9600
Confusion Matrix: [[36 1] [ 2 74]]
```

Figure 4: Logistic Regression terminal output and Confusion Matrix.

The matrix shows that the model correctly classified 36 True Negatives and 74 True Positives, while producing only 1 False Positive and 2 False Negatives.

5.3 Model Comparison

A summary chart was built to easily track the differences between accuracy and F1 scores across our test run models.

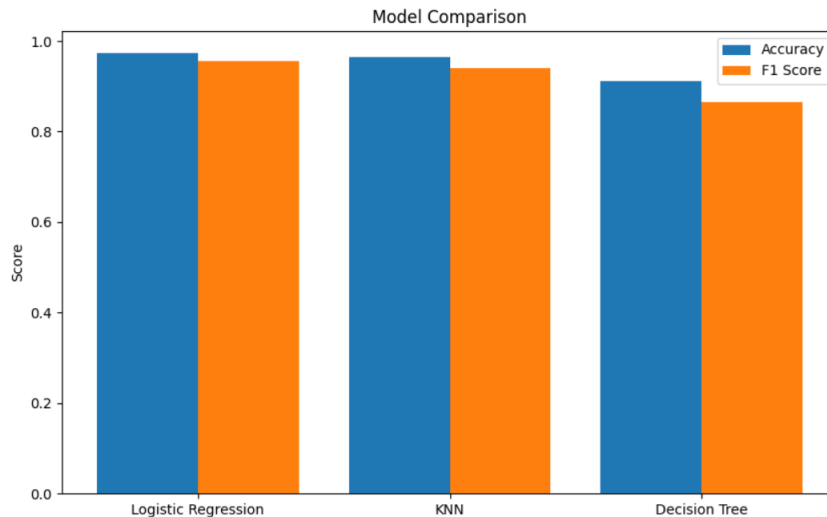


Figure 5: Comparison of Accuracy and F1 Score across all models.

6 Challenges Encountered and Solutions

Understanding the Algorithms: Initially, understanding the mathematical concepts behind gradient descent, entropy, and information gain was challenging.

From-Scratch Implementation: Implementing algorithms without standard frameworks required translating mathematical formulas into clean Python expressions.

Debugging: Shape mismatches between matrix dimensions during dot products, and handling division by zero errors when validation subsets yielded no true/false positive flags were handled through isolated unit testing.

Performance Optimization: Python and NumPy vectorized operations such as `np.dot()`, `np.mean()`, `np.sum()`, and logical array masking were used whenever possible to improve computational efficiency.

7 Limitations

- **Dataset:** Limited dataset size (569 records); results may not generalize perfectly to unseen clinical populations.
- **Models:** Logistic Regression assumes linear decision boundaries; KNN can become computationally expensive; Decision Trees are highly sensitive to training data variations.
- **Training:** Hyperparameter tuning was limited, and cross-validation was not fully integrated into the final evaluation pipeline.

8 Conclusion

This project successfully demonstrated the implementation of several machine learning algorithms from scratch for breast cancer diagnostic classification. The experiments showed that Logistic Regression achieved the best overall performance on the dataset. The project also confirmed that a well-designed machine learning pipeline can be built without relying heavily on high-level libraries. Understanding the mathematical and computational foundations of ML models is essential for effectively applying them to real-world medical problems.